Title: An open-source, provenance-aware framework for reconstructing missing CT dose indices from DICOM metadata

Running title: Provenance-aware CT dose reconstruction

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Conflicts of interest. The author is the representative of LISIT Co., Ltd. (Tokyo, Japan) and Chief Executive Officer of TexelCraft OÜ, and this work was conducted under the supervision of LISIT Co., Ltd. The software is released under the MIT License and the author has a commercial interest in downstream products that may use it. No patient or customer data were used.

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Ethics approval and consent. Not applicable. This study used only publicly available, de-identified

imaging data (TCIA collections released under Creative Commons licenses) and third-party sample files. No institutional review board approval and no informed consent were required.

Data and code availability. The software is available at https://github.com/Institute-of-One/ctdose-core under the MIT License. The version used in this study (version 0.2.1) is archived at Zenodo (https://doi.org/10.5281/zenodo.24081544).

Supplementary Material. One supplementary file accompanies this Technical Note, containing the verification of the size layer and of the RDSR parser, the auditable diagnostic-reference-level comparison, the sample licensing and provenance, the scanner models the coefficient table does not cover, the stratified accuracy detail, the size-correction and effective-dose results, the data-quality findings, and the GE validation arm in full. It contains no figures.

Use of artificial intelligence. A generative AI assistant (Claude, Anthropic) supported software development and drafted portions of this manuscript. All results were produced by executable code in the released repository and were re-run and verified by the author, who takes full responsibility for this work. The AI system is not an author and does not meet authorship criteria.

Author contributions. S.Y.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing — original draft, writing — review and editing, visualization.